

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I, Lakshna sree T.S. (192524059) certify that this submission is my original work and that I [] I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Assessment tool 3 - CO-1

Short Answer test

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course name: cloud computing and big data Analysis

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Course Code: CSA1507

1. Define cloud computing and explain any four characteristics that makes it different from tradition on-premise computing.

cloud computing:

cloud computing provides the delivery of services such as storage, server, internet requirement.

- * It is accessible widely or globally.

- * It provides services over internet.

- * Companies like Amazon, Microsoft developed their own cloud resources including google they are Amazon EC2, Azure and google cloud.

Characteristics of cloud computing:

- * On-demand self service:

User can access the cloud resource without human intervention or they doesnot require an external IT staff for approval. Whereas on traditional computing installation and physical procurement are required.

- * Rapid elasticity:

Resources can be scaled up or down based on the workload but in traditional computing resources are rigid.

* Resource pooling:

Resources can be shared based on the requirement.

* Broad Network Access:

Network can be accessed through devices such as laptops, computers and other devices.

2) Trace the evolution of cloud computing from hardware evolution to internet software evolution and show how these changes enabled cloud computing.

Evolution of cloud computing:

a) Hardware evolution

In 1950s, mainframe systems are developed and later 1990s Personal computers were introduced.

^{Network}
b) ~~Internet~~ evolution:

Then networks like Local Area Network (LAN), and Wide Area Network (WAN) were introduced that helped in overall communication through network from different and wider regions.

c) & Internet evolution:

Next the internet were introduced through which they were accessible through mobile phones and computers.

d) Software evolution:

In this the desktop applications that was used earlier then converted to web applications.

e) Virtualization:

Later virtualization were introduced.

Virtualization means in one computer more than one virtual machines can run.

d) Modern cloud computing:

Companies like Amazon, google, Microsoft developed cloud such as AWS EC2, google cloud, Azure. As cloud computing helps in providing services such as storage, servers, etc.

3) Explain server virtualization. Differentiate server virtualization and cloud computing.

Server virtualization

Server virtualization means creating more than one virtual server in a physical server using hypervisor.

It shares resources with cloud computing.

characteristics:

It Maximizes the hardware utilization

It support resource sharing/pooling.

It is highly reliable and scalable.

It backups every data of the user and helps in easy data recovery.

virtualization	cloud computing
* It helps in creating more than one virtual machine.	* It provide services such as storage, server, network
* They helps in resource optimization.	* They help in service providing.
* They are not accessi-ble widely.	* They are accessible widely.
* eg: VM-ware	* eg: Azure

4) Data centers in delivering cloud services

Data centers

- * Data center provides facilities such as storage server, large amount of data.
- * This data center acts as backbone of cloud services.
- * Data centers helps in storing large amount of user data.
- * The data center has ~~security~~⁴ such as firewall that helps in securing users data.
- * Data centers are highly reliable and scalable.
- * In cloud services, ~~data~~ the user datas are stored in data centers.
- * Data centers are highly scalable.

* Data centre has backup where all the data are backed up repeatedly and data are easy for recovering it.

* All data of users are stored in data centres. It acts as backbone of cloud computing.

5) IaaS, PaaS, SaaS, XaaS

Model	Explanation	Examples.
IaaS	Infrastructure as a Service. In this service model user handles the operating system and the hardware application.	Amazon EC2
PaaS	Provider as a Service In this service model the developer focuses only on the coding rather than infrastructure.	Google Engine
SaaS	Software as a Service No installation is required. They are ready to use web application through web browser.	Gmail.

XaaS

Everything as a
Service

This model provides
all services such as
Storage as a service
and database as a
service

Microsoft 365